

LATAM Bank

Complete Data Dictionary

Factored Datathon 2026

Dataset Version 1.0.0

Total Records	~19,000,000
Tables	13
Countries	Mexico, Colombia, Argentina
Date Range	June 2023 - June 2026
Currencies	MXN, COP, ARS, USD

Generated: September 2026

Data Access Credentials

The dataset is hosted on **Amazon S3** with **read-only** access for Datathon participants.

Attribute	Value
Bucket Name	factored-datathon-2026-s3-157725502942-us-east-2-an
Region	us-east-2
Access Key ID	AKIASJOJMFEXPDNU7JH7C
Secret Access Key	O43PZL4gzfoQDm8Xk6qDxWidJmYGsRjaTYd5eO1F
Permissions	Read-Only

AWS CLI Configuration

Configure your AWS CLI with the following commands:

```
aws configure set aws_access_key_id AKIASJOJMFEXPDNU7JH7C
aws configure set aws_secret_access_key O43PZL4gzfoQDm8Xk6qDxWidJmYGsRjaTYd5eO1F
aws configure set region us-east-2
```

Common AWS S3 Commands

List all available data files:

```
aws s3 ls s3://factored-datathon-2026-s3-157725502942-us-east-2-an/data/
```

Download a specific file:

```
aws s3 cp s3://factored-datathon-2026-s3-157725502942-us-east-2-an/data/customers.csv ./
```

Sync entire data folder:

```
aws s3 sync s3://factored-datathon-2026-s3-157725502942-us-east-2-an/data/ ./data/
```

■ **Important:** These credentials provide **read-only** access to the dataset. Do not share these credentials outside of Datathon participants.

Dataset Overview

This dataset simulates a **regional banking system** operating across three Latin American countries: Mexico, Colombia, and Argentina. It contains realistic banking, customer service, and digital interaction data spanning **three years** (June 17, 2023 to June 17, 2026).

The dataset includes **19,000,000 records** distributed across **13 tables**, covering customer master data, financial transactions, call center interactions, marketing campaigns, customer complaints, and digital events.

Languages: All text data is in Spanish with regional variations (Mexican, Colombian, Argentine accents).

Currencies: MXN, COP, ARS, USD

Data Generation: Synthetically generated for the Factored Datathon 2026.

Data Quality Characteristics

This dataset intentionally includes real-world data quality challenges:

- **Duplicate Records:** ~2% across tables
- **Null Values:** ~5% in nullable fields
- **Late Arrivals:** Some partitioned data may arrive late
- **Schema Evolution:** Table schemas may change over time

Dimension Tables

Reference data tables containing 5 dimension tables.

Customers [DIMENSION]

Rows: 150,000 | Source: Core Banking | Partition: monthly_snapshot

Bank customer dimension table

Column	Type	Description	Constraints
customer_id	VARCHAR(20)	Unique customer ID	PK, NOT NULL
document_number	VARCHAR(20)	Identity document number	NOT NULL, UNIQUE
document_type	VARCHAR(10)	Document type (DNI, CURP, CC, CE, Passport)	NOT NULL
first_name	VARCHAR(100)	Customer first name (in Spanish)	NOT NULL
last_name	VARCHAR(100)	Customer last name (in Spanish)	NOT NULL
date_of_birth	DATE	Date of birth	NOT NULL
gender	VARCHAR(1)	Gender (M, F, O)	-
email	VARCHAR(100)	Email address	-
mobile_phone	VARCHAR(20)	Mobile phone number	-
landline_phone	VARCHAR(20)	Landline phone number	-
address	VARCHAR(200)	Full address (in Spanish)	-
city	VARCHAR(100)	City of residence	NOT NULL
state	VARCHAR(100)	State/Province	NOT NULL
country	VARCHAR(50)	Country (Mexico, Colombia, Argentina)	NOT NULL
postal_code	VARCHAR(10)	Postal code	-
detected_accent	VARCHAR(50)	Spanish accent detected (mexican, colombian, argentine, neutral)	
segment	VARCHAR(50)	Customer segment (Premium, Plus, Basic, Student)	NOT NULL
credit_score	INTEGER	Credit score (300-850)	-
estimated_monthly_income	DECIMAL(12,2)	Estimated monthly income in local currency	-
occupation	VARCHAR(100)	Customer occupation	-
marital_status	VARCHAR(20)	Marital status	-
education_level	VARCHAR(50)	Education level	-
registration_date	TIMESTAMP	Registration date as customer	NOT NULL
registration_branch_id	VARCHAR(20)	Branch ID where registered	FK, NOT NULL
customer_status	VARCHAR(20)	Status (Active, Inactive, Suspended, Closed)	NOT NULL
last_updated	TIMESTAMP	Last record update	NOT NULL

accepts_marketing	BOOLEAN	Accepts marketing communications	NOT NULL
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Products [DIMENSION]

Rows: 400,000 | Source: Core Banking | Partition: monthly_snapshot

Active financial products of customers

Column	Type	Description	Constraints
product_id	VARCHAR(20)	Unique product ID	PK, NOT NULL
customer_id	VARCHAR(20)	Owner customer ID	FK, NOT NULL
product_type	VARCHAR(50)	Product type (Checking Account, Savings Account, Credit Card, Debit Card, Personal Loan, Mortgage, Investment)	NOT NULL
product_number	VARCHAR(30)	Account/card/policy number	NOT NULL, UNIQUE
currency	VARCHAR(3)	Currency (MXN, COP, ARS, USD)	NOT NULL
current_balance	DECIMAL(15,2)	Current balance	NOT NULL
credit_limit	DECIMAL(15,2)	Credit limit (for credit products)	-
interest_rate	DECIMAL(5,2)	Annual interest rate (%)	-
opening_date	DATE	Product opening date	NOT NULL
expiration_date	DATE	Expiration date (for term products)	-
opening_branch_id	VARCHAR(20)	Branch where opened	FK, NOT NULL
product_status	VARCHAR(20)	Status (Active, Blocked, Closed, Suspended)	NOT NULL
opening_channel	VARCHAR(30)	Opening channel (Branch, Web, App, Call Center)	NOT NULL
has_linked_app	BOOLEAN	Product linked to mobile app	NOT NULL
days_past_due	INTEGER	Days past due (for credits)	-
last_transaction_date	TIMESTAMP	Last transaction date	-
last_updated	TIMESTAMP	Last record update	NOT NULL

Branches [DIMENSION]

Rows: 350 | Source: Internal | Partition: full_snapshot

Physical bank branches

Column	Type	Description	Constraints
branch_id	VARCHAR(20)	Unique branch ID	PK, NOT NULL
branch_code	VARCHAR(10)	Internal branch code	NOT NULL, UNIQUE
branch_name	VARCHAR(100)	Branch name	NOT NULL
branch_type	VARCHAR(30)	Type (Main, Express, Premium, Corporate)	NOT NULL

address	VARCHAR(200)	Full address (in Spanish)	NOT NULL
city	VARCHAR(100)	City	NOT NULL
state	VARCHAR(100)	State/Province	NOT NULL
country	VARCHAR(50)	Country	NOT NULL
postal_code	VARCHAR(10)	Postal code	-
geographic_zone	VARCHAR(50)	Zone (Urban, Suburban, Rural)	NOT NULL
phone	VARCHAR(20)	Contact phone	NOT NULL
email	VARCHAR(100)	Branch email	-
opening_time	TIME	Opening time	NOT NULL
closing_time	TIME	Closing time	NOT NULL
has_atms	BOOLEAN	Has ATMs	NOT NULL
atm_count	INTEGER	Number of ATMs	-
has_teller_windows	BOOLEAN	Has teller windows	NOT NULL
teller_window_count	INTEGER	Number of teller windows	-
latitude	DECIMAL(10,7)	Geographic latitude	-
longitude	DECIMAL(10,7)	Geographic longitude	-
branch_opening_date	DATE	Branch opening date	NOT NULL
branch_status	VARCHAR(20)	Status (Active, Temporarily Closed, Closed)	NOT NULL

Service Agents [DIMENSION]

Rows: 1,200 | Source: Internal | Partition: monthly_snapshot

Customer service agents

Column	Type	Description	Constraints
agent_id	VARCHAR(20)	Unique agent ID	PK, NOT NULL
employee_code	VARCHAR(15)	Employee code	NOT NULL, UNIQUE
first_name	VARCHAR(100)	Agent first name (in Spanish)	NOT NULL
last_name	VARCHAR(100)	Agent last name (in Spanish)	NOT NULL
email	VARCHAR(100)	Corporate email	NOT NULL
phone	VARCHAR(20)	Contact phone	-
native_accent	VARCHAR(50)	Native Spanish accent (mexican, colombian, argentine)	NOT NULL
country_of_origin	VARCHAR(50)	Country of origin	NOT NULL
assigned_branch_id	VARCHAR(20)	Assigned branch	FK
agent_type	VARCHAR(30)	Type (Phone, In-Person, Digital, Hybrid)	NOT NULL

experience_level	VARCHAR(20)	Level (Junior, Mid-Senior, Senior, Specialist)	NOT NULL
languages	VARCHAR(100)	Languages spoken	NOT NULL
specialty	VARCHAR(100)	Specialty	-
hire_date	DATE	Hire date	NOT NULL
avg_csat	DECIMAL(3,2)	Average CSAT score (1-5)	-
total_monthly_interactions	INTEGER	Total interactions in last month	-
agent_status	VARCHAR(20)	Status (Active, Vacation, Leave, Inactive)	NOT NULL
work_shift	VARCHAR(20)	Shift (Morning, Afternoon, Night, Rotating)	NOT NULL

Marketing Campaigns [DIMENSION]

Rows: 200 | Source: Internal | Partition: full_snapshot

Bank marketing campaigns

Column	Type	Description	Constraints
campaign_id	VARCHAR(20)	Unique campaign ID	PK, NOT NULL
campaign_name	VARCHAR(150)	Campaign name	NOT NULL
description	TEXT	Campaign description	-
campaign_type	VARCHAR(50)	Type (Email, SMS, Push, WhatsApp, Voice, Mix)	NOT NULL
campaign_objective	VARCHAR(100)	Objective (Acquisition, Retention, Cross-sell, Up-sell, Reactivation)	NOT NULL
promoted_product	VARCHAR(50)	Promoted product	-
target_segment	VARCHAR(50)	Target segment	-
target_country	VARCHAR(50)	Target country	-
start_date	DATE	Start date	NOT NULL
end_date	DATE	End date	NOT NULL
budget	DECIMAL(12,2)	Assigned budget	-
campaign_status	VARCHAR(20)	Status (Planned, Active, Paused, Completed)	NOT NULL
expected_conversion_rate	DECIMAL(5,2)	Expected conversion rate (%)	-

Fact Tables

Transactional and event data tables containing 7 fact tables.

Transactions [FACT]

Rows: 5,000,000 | Source: Core Banking | Partition: daily

Daily financial transactions

Column	Type	Description	Constraints
transaction_id	VARCHAR(30)	Unique transaction ID	PK, NOT NULL
transaction_date	TIMESTAMP	Transaction date and time	NOT NULL
process_date	DATE	Process date (partition key)	NOT NULL
product_id	VARCHAR(20)	Product ID	FK, NOT NULL
customer_id	VARCHAR(20)	Customer ID	FK, NOT NULL
transaction_type	VARCHAR(50)	Type (Deposit, Withdrawal, Transfer, Payment, Purchase, Adjustment)	NOT NULL
transaction_category	VARCHAR(50)	Category (Food, Transport, Services, Entertainment, Health, Other)	NOT NULL
amount	DECIMAL(15,2)	Transaction amount	NOT NULL
currency	VARCHAR(3)	Currency	NOT NULL
amount_usd	DECIMAL(15,2)	Amount converted to USD	-
channel	VARCHAR(30)	Channel (ATM, Branch, Web, App, POS, Transfer)	NOT NULL
branch_id	VARCHAR(20)	Branch ID (if applicable)	FK
merchant_name	VARCHAR(150)	Merchant name (for purchases)	-
merchant_category	VARCHAR(50)	MCC merchant category	-
transaction_country	VARCHAR(50)	Country where transaction occurred	NOT NULL
transaction_city	VARCHAR(100)	City where transaction occurred	-
transaction_status	VARCHAR(20)	Status (Approved, Declined, Pending, Reversed)	NOT NULL
response_code	VARCHAR(10)	System response code	-
is_fraud	BOOLEAN	Marked as fraud	NOT NULL
fraud_score	DECIMAL(5,2)	Fraud risk score (0-100)	-
latitude	DECIMAL(10,7)	Transaction latitude	-
longitude	DECIMAL(10,7)	Transaction longitude	-

Call Center Interactions [FACT]

Rows: 800,000 | Source: Contact Center | Partition: daily

Call center interactions with customers

Column	Type	Description	Constraints
interaction_id	VARCHAR(30)	Unique interaction ID	PK, NOT NULL
interaction_date	TIMESTAMP	Interaction date and time	NOT NULL
process_date	DATE	Process date (partition key)	NOT NULL
customer_id	VARCHAR(20)	Customer ID	FK, NOT NULL
agent_id	VARCHAR(20)	Agent ID who attended	FK
interaction_type	VARCHAR(30)	Type (Inbound Call, Outbound Call, Chat, Email, Video)	NOT NULL
channel	VARCHAR(30)	Channel (Phone, Web Chat, WhatsApp, Email, App)	NOT NULL
contact_reason	VARCHAR(100)	Main contact reason	NOT NULL
reason_category	VARCHAR(50)	Category (Transactional, Product, Technical, Commercial, Complaint)	NOT NULL
duration_seconds	INTEGER	Duration in seconds	-
wait_time_seconds	INTEGER	Wait time before service	-
was_resolved	BOOLEAN	Resolved on first call (FCR)	-
requires_followup	BOOLEAN	Requires follow-up	NOT NULL
detected_sentiment	VARCHAR(20)	Sentiment (Positive, Neutral, Negative, Very Negative)	-
sentiment_score	DECIMAL(3,2)	Sentiment score (-1 to 1)	-
customer_detected_accent	VARCHAR(50)	Customer's detected accent	-
agent_used_accent	VARCHAR(50)	Accent used by agent in response	-
was_escalated	BOOLEAN	Was escalated to supervisor	NOT NULL
mentioned_products	VARCHAR(200)	Product IDs mentioned (comma-separated)	-
has_transcript	BOOLEAN	Has transcript available	NOT NULL
has_recording	BOOLEAN	Has audio recording	NOT NULL

Call Transcripts [FACT]

Rows: 200,000 | Source: Contact Center | Partition: daily

Call center call transcripts

Column	Type	Description	Constraints
transcript_id	VARCHAR(30)	Unique transcript ID	PK, NOT NULL
interaction_id	VARCHAR(30)	Related interaction ID	FK, NOT NULL
process_date	DATE	Process date (partition key)	NOT NULL
customer_id	VARCHAR(20)	Customer ID	FK, NOT NULL

agent_id	VARCHAR(20)	Agent ID	FK, NOT NULL
full_text	TEXT	Full call transcript (in Spanish)	NOT NULL
customer_text	TEXT	Only what customer said (in Spanish)	-
agent_text	TEXT	Only what agent said (in Spanish)	-
detected_language	VARCHAR(10)	Main language detected	NOT NULL
detected_accent	VARCHAR(50)	Detected accent	-
accent_confidence	DECIMAL(3,2)	Accent detection confidence (0-1)	-
detected_keywords	VARCHAR(500)	Identified keywords	-
mentioned_entities	TEXT	Extracted entities (JSON)	-
detected_intents	VARCHAR(300)	Identified intents	-
main_topics	VARCHAR(300)	Main conversation topics	-
transcription_model	VARCHAR(50)	Model used (Whisper, Google STT, etc.)	NOT NULL
audio_quality	VARCHAR(20)	Audio quality (High, Medium, Low)	-
duration_seconds	INTEGER	Call duration	NOT NULL

Satisfaction Surveys [FACT]

Rows: 250,000 | Source: Contact Center | Partition: daily

Post-interaction satisfaction surveys (CSAT, NPS)

Column	Type	Description	Constraints
survey_id	VARCHAR(30)	Unique survey ID	PK, NOT NULL
survey_date	TIMESTAMP	Response date and time	NOT NULL
process_date	DATE	Process date (partition key)	NOT NULL
interaction_id	VARCHAR(30)	Evaluated interaction ID	FK
customer_id	VARCHAR(20)	Customer ID	FK, NOT NULL
agent_id	VARCHAR(20)	Evaluated agent ID	FK
survey_type	VARCHAR(20)	Type (CSAT, NPS, CES)	NOT NULL
send_channel	VARCHAR(30)	Send channel (Email, SMS, IVR, App, Web)	NOT NULL
main_score	INTEGER	Main score (1-5 for CSAT, 0-10 for NPS)	NOT NULL
nps_category	VARCHAR(20)	NPS category (Promoter, Passive, Detractor)	-
question_1_text	TEXT	Question 1 text	-
question_1_response	INTEGER	Question 1 response (1-5)	-
question_2_text	TEXT	Question 2 text	-
question_2_response	INTEGER	Question 2 response (1-5)	-

question_3_text	TEXT	Question 3 text	-
question_3_response	INTEGER	Question 3 response (1-5)	-
open_comments	TEXT	Customer comments (in Spanish)	-
comment_sentiment	VARCHAR(20)	Comment sentiment	-
response_time_hours	DECIMAL(8,2)	Hours between interaction and response	-
campaign_response_rate	DECIMAL(5,2)	Campaign response rate (%)	-

Digital Events [FACT]

Rows: 10,000,000 | Source: Digital Banking | Partition: daily

Digital channel interaction events (mobile app, web)

Column	Type	Description	Constraints
event_id	VARCHAR(30)	Unique event ID	PK, NOT NULL
event_date	TIMESTAMP	Event date and time	NOT NULL
process_date	DATE	Process date (partition key)	NOT NULL
customer_id	VARCHAR(20)	Customer ID	FK
session_id	VARCHAR(50)	User session ID	NOT NULL
event_type	VARCHAR(50)	Type (PageView, Click, FormSubmit, Login, Logout, Error, Purchase)	NOT NULL
event_category	VARCHAR(50)	Category (Navigation, Transaction, Authentication, Product)	NOT NULL
channel	VARCHAR(30)	Channel (Android App, iOS App, Desktop Web, Mobile Web)	NOT NULL
platform	VARCHAR(30)	Platform (Android, iOS, Windows, MacOS, Linux)	-
browser	VARCHAR(50)	Browser used	-
app_version	VARCHAR(20)	App version	-
page_url	VARCHAR(300)	Page URL	-
page_title	VARCHAR(200)	Page title	-
action	VARCHAR(100)	Action performed	-
element_id	VARCHAR(100)	Interacted element ID	-
product_id	VARCHAR(20)	Related product ID	FK
event_value	DECIMAL(15,2)	Event monetary value (if applicable)	-
duration_seconds	INTEGER	Event duration	-
ip_address	VARCHAR(45)	User IP address	-
ip_country	VARCHAR(50)	Country detected by IP	-
ip_city	VARCHAR(100)	City detected by IP	-
is_mobile	BOOLEAN	Event from mobile device	NOT NULL

referrer	VARCHAR(300)	Referrer URL	-
utm_source	VARCHAR(100)	UTM source	-
utm_medium	VARCHAR(100)	UTM medium	-
utm_campaign	VARCHAR(100)	UTM campaign	-

Complaints [FACT]

Rows: 80,000 | Source: PQR | Partition: daily

Complaints and claims system (PQR)

Column	Type	Description	Constraints
complaint_id	VARCHAR(30)	Unique complaint/claim ID	PK, NOT NULL
creation_date	TIMESTAMP	Complaint creation date	NOT NULL
process_date	DATE	Process date (partition key)	NOT NULL
customer_id	VARCHAR(20)	Customer ID	FK, NOT NULL
case_type	VARCHAR(30)	Type (Complaint, Claim, Request, Suggestion)	NOT NULL
category	VARCHAR(100)	Case category	NOT NULL
subcategory	VARCHAR(100)	Subcategory	-
reception_channel	VARCHAR(30)	Channel (Call Center, Email, Web, App, Branch, Regulator)	NOT NULL
affected_product_id	VARCHAR(20)	Affected product ID	FK
related_branch_id	VARCHAR(20)	Related branch ID	FK
origin_interaction_id	VARCHAR(30)	Originating interaction ID	FK
description	TEXT	Case description (in Spanish)	NOT NULL
claimed_amount	DECIMAL(15,2)	Claimed amount (if applicable)	-
currency	VARCHAR(3)	Claimed amount currency	-
priority	VARCHAR(20)	Priority (Low, Medium, High, Critical)	NOT NULL
status	VARCHAR(30)	Status (Open, In Process, Escalated, Resolved, Closed, Rejected)	NOT NULL
assigned_agent_id	VARCHAR(20)	Assigned agent ID	FK
assignment_date	TIMESTAMP	Assignment date	-
first_response_date	TIMESTAMP	First response date	-
resolution_date	TIMESTAMP	Resolution date	-
closing_date	TIMESTAMP	Closing date	-
sla_breached	BOOLEAN	SLA breached	NOT NULL
resolution_days	INTEGER	Days to resolution	-
resolution	TEXT	Resolution description (in Spanish)	-

compensation_granted	DECIMAL(15,2)	Compensation amount granted	-
resolution_satisfaction	INTEGER	Resolution satisfaction score (1-5)	-
is_repeat_complainer	BOOLEAN	Customer with previous complaints in last 90 days	NOT NULL

Campaign Sends [FACT]

Rows: 2,000,000 | Source: Internal | Partition: daily

Individual marketing campaign sends

Column	Type	Description	Constraints
send_id	VARCHAR(30)	Unique send ID	PK, NOT NULL
send_date	TIMESTAMP	Send date and time	NOT NULL
process_date	DATE	Process date (partition key)	NOT NULL
campaign_id	VARCHAR(20)	Campaign ID	FK, NOT NULL
customer_id	VARCHAR(20)	Recipient customer ID	FK, NOT NULL
send_channel	VARCHAR(30)	Channel (Email, SMS, Push, WhatsApp, Voice)	NOT NULL
template_used	VARCHAR(100)	Template used	-
subject	VARCHAR(200)	Message subject	-
send_status	VARCHAR(20)	Status (Sent, Failed, Bounced, Blocked)	NOT NULL
was_delivered	BOOLEAN	Was delivered successfully	NOT NULL
was_opened	BOOLEAN	Was opened/read	-
open_date	TIMESTAMP	Open date	-
was_clicked	BOOLEAN	Clicked on any link	-
click_date	TIMESTAMP	First click date	-
click_count	INTEGER	Total click count	-
had_conversion	BOOLEAN	Converted (completed desired action)	NOT NULL
conversion_date	TIMESTAMP	Conversion date	-
conversion_value	DECIMAL(15,2)	Conversion monetary value	-
open_device	VARCHAR(30)	Device used to open	-
open_country	VARCHAR(50)	Country where opened	-
failure_reason	VARCHAR(200)	Failure reason (if applicable)	-
send_cost	DECIMAL(10,4)	Individual send cost	-

Reference Tables

Reference data for lookups and conversions.

Daily Exchange Rates [REFERENCE]

Rows: 3,000 | Source: Reference | Partition: daily

Daily exchange rates for currency conversion

Column	Type	Description	Constraints
date	DATE	Exchange rate date	PK, NOT NULL
source_currency	VARCHAR(3)	Source currency	PK, NOT NULL
target_currency	VARCHAR(3)	Target currency	PK, NOT NULL
exchange_rate	DECIMAL(12,6)	Exchange rate	NOT NULL
buy_rate	DECIMAL(12,6)	Bank buy rate	-
sell_rate	DECIMAL(12,6)	Bank sell rate	-
source	VARCHAR(50)	Exchange rate source	-

Foreign Key Relationships

The following foreign key relationships exist between tables:

customers

- products.customer_id → customers.customer_id
- transactions.customer_id → customers.customer_id
- call_center_interactions.customer_id → customers.customer_id
- call_transcripts.customer_id → customers.customer_id
- satisfaction_surveys.customer_id → customers.customer_id
- digital_events.customer_id → customers.customer_id
- complaints.customer_id → customers.customer_id
- campaign_sends.customer_id → customers.customer_id

branches

- customers.registration_branch_id → branches.branch_id
- products.opening_branch_id → branches.branch_id
- service_agents.assigned_branch_id → branches.branch_id
- transactions.branch_id → branches.branch_id
- complaints.related_branch_id → branches.branch_id

service_agents

- call_center_interactions.agent_id → service_agents.agent_id
- call_transcripts.agent_id → service_agents.agent_id
- satisfaction_surveys.agent_id → service_agents.agent_id
- complaints.assigned_agent_id → service_agents.agent_id

products

- transactions.product_id → products.product_id
- digital_events.product_id → products.product_id
- complaints.affected_product_id → products.product_id

marketing_campaigns

- campaign_sends.campaign_id → marketing_campaigns.campaign_id

call_center_interactions

- call_transcripts.interaction_id → call_center_interactions.interaction_id

- satisfaction_surveys.interaction_id → call_center_interactions.interaction_id
- complaints.origin_interaction_id → call_center_interactions.interaction_id

Potential Use Cases

Customer Analytics

- Customer segmentation and clustering
- Churn prediction models
- Customer lifetime value (CLV) analysis
- Cross-sell and up-sell opportunity identification

Contact Center Optimization

- First Call Resolution (FCR) improvement
- Agent performance analysis
- Sentiment trend analysis
- Accent-based routing optimization

Fraud Detection

- Transaction fraud detection models
- Anomaly detection in spending patterns
- Geographic risk modeling

Marketing Analytics

- Campaign effectiveness measurement
- Channel attribution modeling
- Personalization models
- A/B testing analysis

NLP / Text Analytics

- Topic modeling on call transcripts
- Intent classification
- Entity extraction
- Multilingual accent detection

Important Notes

- **Spanish Language Data:** All text data is in Spanish with regional variations.
- **Accent Detection:** Dataset includes Mexican, Colombian, and Argentine accent fields.
- **Multi-Currency:** Transactions include local currency and USD conversion.
- **Date Partitioning:** Large fact tables partitioned by year/month/day.
- **Synthetic Data:** Completely synthetic - no real customer information.
- **Referential Integrity:** FK relationships maintained (small % of orphans for testing).

Factored Datathon 2026

Dataset Version 1.0.0

For questions, contact the Factored Datathon team